Metals, acids, salts and chemical tests

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

1 · Part 2

Paper 2 only | Extraction and uses of metals | [1]

Answer: C

Worked solution

The less reactive a metal is, the more likely it is to be found as the element itself. Gold is near the bottom of the reactivity series, so it can be dug out of the ground as the metal. Aluminium, iron and zinc are too reactive and occur as oxides or other compounds that must be extracted.

2 · Part 2

Paper 2 only | Extraction and uses of metals | [1]

Answer: A

Worked solution

Hardness depends on how easily the layers of atoms can slide over each other. Adding atoms of another element, which are a different size, breaks up the regular pattern. The layers can no longer slide past each other easily, so the alloy is harder than the pure metal.

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3 · Part 2

Paper 2 only | Extraction and uses of metals | [7]

Mark scheme

(a) **[2]**

- carbon is more reactive than iron OR carbon is above iron in the reactivity series (1)
- so carbon can remove oxygen from iron(III) oxide OR carbon reduces iron(III) oxide (1)

ALLOW carbon displaces iron from its oxide

(b) **[2]**

- iron(III) oxide OR Fe_2O_3 (1)
- it loses oxygen (1)

ALLOW iron(III) ions gain electrons for the second mark

REJECT carbon monoxide OR CO (scores 0 for both marks)

- the second mark depends on the correct substance

(c) **[2]**

- aluminium is more reactive than carbon OR aluminium is above carbon in the reactivity series (1)
- so carbon cannot remove oxygen from aluminium oxide OR carbon cannot reduce aluminium oxide (1)

(d) **[1]**

- gold is very unreactive, so it is found as the uncombined element OR as the metal itself (1)

ALLOW gold is found native

Worked solution

(a) A more reactive element can take oxygen away from the oxide of a less reactive metal. Carbon is above iron in the series, so carbon removes the oxygen from iron(III) oxide and leaves iron.

(b) Reduction is the loss of oxygen. In the equation, Fe_2O_3 loses its oxygen to become Fe, so iron(III) oxide is reduced. Carbon monoxide gains oxygen to become CO_2, so it is oxidised.

(c) Aluminium is above carbon in the reactivity series. Carbon is not reactive enough to take oxygen from aluminium oxide, so a more powerful method, electrolysis, must be used.

(d) Gold is so unreactive that it is found in the Earth's crust as the metal, not in a compound, so it does not need to be extracted chemically.

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4 · Part 2

Paper 2 only | Extraction and uses of metals | [8]

Mark scheme

(a) **[1]**

- $TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$ (both 2s correct) (1)

(b) **[2]**

- titanium is less reactive than magnesium (1)
- because magnesium displaces titanium from titanium(IV) chloride OR magnesium takes the chlorine from the titanium (1)
- the second mark depends on the first

(c) **[2]**

- argon is unreactive OR inert (1)
- in air the hot magnesium OR titanium would react with oxygen (or nitrogen) OR would form oxides OR would burn (1)

(d) **[3]**

Indicative points, any three (1 each):

- it is a batch process, not continuous, so less titanium is made in a given time (1)
- Stage 3 is slow OR takes several days (1)
- there are many stages (four), each adding cost (1)
- high temperatures (about 900 °C) need a lot of energy OR fuel (1)
- the magnesium is made by electrolysis, which uses a lot of electricity (1)
- the reactants magnesium, chlorine and argon are expensive compared with the carbon used for iron (1)

ALLOW iron is cheaper because it is made continuously using cheap carbon, as a comparison point for one mark

Worked solution

(a) Each $TiCl_4$ has four chlorine atoms. Each $MgCl_2$ has two, so $2MgCl_2$ are needed, which uses 2Mg: $TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$.

(b) Magnesium takes the chlorine from titanium and leaves titanium metal. This is a displacement reaction, which only happens when the free metal is more reactive. So titanium is less reactive than magnesium.

(c) At 900 °C, magnesium and titanium would react with the oxygen in air and form oxides, so no pure titanium would be made. Argon is unreactive, so it protects the metals.

(d) Iron is made all the time in one continuous process using cheap carbon. Titanium is made in batches, Stage 3 alone takes days, there are four stages, the temperatures are high, and the magnesium has to be made by electrolysis, which uses a lot of electricity. All of these add to the cost, so titanium is expensive.

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5 · Part 2

Paper 2 only | Extraction and uses of metals | [6]

Mark scheme

(a) **[4]**

- aluminium: low density OR good conductor of electricity OR resists corrosion (1)

ALLOW lightweight

- copper: good conductor of electricity OR ductile OR can be drawn into wires (1)
- stainless steel: resists corrosion OR does not rust (1)
- high-carbon steel: hard OR strong OR keeps a sharp edge (1)

IGNORE cheap OR shiny on their own

(b) **[2]**

- mild steel is softer OR more malleable, so it can be pressed or bent into shape (1)
- high-carbon steel is harder but more brittle, so it would crack when shaped (1)

ALLOW mild steel is less brittle than high-carbon steel

Worked solution

(a) Overhead cables must carry electricity and be light enough to hang between pylons, so aluminium's low density and good conductivity matter. House wiring must conduct electricity well and be drawn into thin wires, which copper does. Cutlery and sinks get wet all the time, so stainless steel is used because it does not rust. Cutting tools must be very hard, which is why high-carbon steel is used.

(b) Car body panels are pressed into curved shapes. Mild steel is soft and malleable, so it bends without breaking. High-carbon steel is harder but brittle and would crack.

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6 · Part 2 Paper 2 only Extraction and uses of metals [6]

Mark scheme

(a) **[1]**

- a mixture of a metal with one or more other elements, usually other metals or carbon (1)

> **REJECT** a compound

(b) **[1]**

- as the percentage of zinc increases, the hardness increases (1)

(c) **[3]**

- in pure copper all the atoms are the same size and are arranged in regular layers (1)
- the layers can slide over each other easily (1)
- in brass the zinc atoms are a different size, which disrupts the layers, so the layers cannot slide over each other as easily (1)

(d) **[1]**

- any value from 79 to 82 (1)

Worked solution

(a) An alloy is a mixture of a metal with at least one other element, often another metal or carbon. Brass is copper mixed with zinc.

(b) Each 10% more zinc raises the hardness number by about 12, so more zinc makes harder brass.

(c) Pure copper has identical atoms packed in neat layers, which slide over each other easily when a force is applied, so it is fairly soft. Zinc atoms are a different size from copper atoms. They disturb the regular layers, so it is harder for the layers to slide, and the brass is harder.

(d) 25% is halfway between 20% and 30%, so the hardness number is about halfway between 75 and 86: (75 + 86) ÷ 2 = 80.5, so about 80.

7 · Part 2 Paper 1 & 2 Acids, alkalis and titrations [1]

Answer: D

Worked solution

Use the pH bands: 0–3 strongly acidic, 4–6 weakly acidic, 7 neutral, 8–10 weakly alkaline, 11–14 strongly alkaline. 9 lies in the 8–10 band.

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8 · Part 2

Paper 1 & 2 | Acids, alkalis and titrations | [1]

Answer: B

Worked solution

Remember each indicator's acid and alkali colours. Litmus: red in acid, blue in alkali. Phenolphthalein: colourless in acid, pink in alkali. Methyl orange: red in acid, yellow in alkali. Sodium hydroxide solution is an alkali, so the row is blue, pink, yellow.

9 · Part 2

Paper 1 & 2 | Acids, alkalis and titrations | [6]

Mark scheme

(a) **[2]**

- put a drop of solution J on the universal indicator paper using a glass rod OR dip the paper into solution J (1)
- compare the colour of the paper with a colour chart (to read off the pH) (1)

(b) **[1]**

- K (1)

(c) **[1]**

- M (1)

(d) **[1]**

- H^+ (1)

ALLOW H^+(aq) OR hydrogen ion

(e) **[1]**

- OH^- (1)

ALLOW OH^-(aq) OR hydroxide ion

Worked solution

(a) Add a little of the solution to the indicator paper, then match the colour against the pH colour chart.

(b) Strongly alkaline is pH 11–14, so K (pH 13).

(c) Weakly acidic is pH 4–6, so M (pH 5). J (pH 1) is strongly acidic.

(d) Acids in water release hydrogen ions, H^+.

(e) Alkalis in water release hydroxide ions, OH^-.

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10 · Part 2

Paper 1 & 2 | Acids, alkalis and titrations | [6]

Mark scheme

(a) **[2]**

- the pH rises only slightly OR stays low (strongly acidic) from 0 to 16 cm^3 (1)
- the pH then rises sharply between 16 and 20 cm^3, then increases only slightly (strongly alkaline, about 12) (1)

(b) **[1]**

- 16 cm^3 and 20 cm^3 (1)

(c) **[2]**

- hydroxide ions (OH^-) from the alkali react with hydrogen ions (H^+) from the acid to form water (1)

ALLOW $H^+ + OH^- \rightarrow H_2O$

- so the concentration of hydrogen ions decreases (1)

ALLOW after neutralisation, there are excess hydroxide ions for the second mark

(d) **[1]**

- add the sodium hydroxide solution in smaller volumes (e.g. 0.5 cm^3 OR drop by drop) between 16 cm^3 and 20 cm^3 and measure the pH after each addition (1)

ALLOW use a burette to add the alkali

ALLOW use an indicator and add the alkali drop by drop until the colour changes

Worked solution

(a) For the first 16 cm^3 the pH barely changes, from 1.0 to 2.2, because there is still plenty of acid. Between 16 and 20 cm^3 it jumps from 2.2 to 11.9, then rises only slowly to 12.4.

(b) pH 7 lies between 2.2 (16 cm^3) and 11.9 (20 cm^3), so the mixture became neutral between 16 and 20 cm^3.

(c) The alkali provides OH^- ions, which react with the H^+ ions from the acid to make water. As H^+ ions are removed, the solution becomes less acidic and the pH goes up. After the acid has been used up, extra OH^- ions make the mixture alkaline.

(d) The 4 cm^3 steps are too large to find the neutral point exactly. Adding small amounts (or drops) in the 16–20 cm^3 region pins it down more precisely.

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11 · Part 2

Paper 2 only | Acids, alkalis and titrations | [7]

Mark scheme

(a) **[5]**

Indicative points, any five (1 each):

- rinse the burette with the acid, then fill it with acid using the funnel (remove the funnel) (1)
- record the initial burette reading (1)
- use the pipette and filler to transfer 25.0 cm^3 of sodium hydroxide solution into the conical flask (1)
- add a few drops of phenolphthalein to the flask (1)
- add acid from the burette while swirling the flask (1)
- add the acid drop by drop near the end point (1)
- stop when the indicator changes from pink to colourless (1)
- record the final burette reading and subtract to find the volume of acid added (1)
- repeat until concordant titres are obtained (within 0.10 cm^3) and calculate the mean (1)

(b) **[1]**

- to see the colour change (end point) clearly (1)

(c) **[1]**

- a pipette measures the volume more accurately OR more precisely (1)

Worked solution

(a) Fill the burette with the acid using the funnel, then take the funnel out and note the starting reading. Use the pipette and filler to put exactly 25.0 cm^3 of sodium hydroxide solution into the conical flask, then add a few drops of phenolphthalein, which turns pink. Run in acid from the burette while swirling. Near the end, add the acid a drop at a time. Stop as soon as the pink colour disappears. Read the burette again and subtract to find the titre. Repeat until two titres agree within 0.10 cm^3 and take the mean.

(b) The white background makes the exact moment of colour change easy to see.

(c) A pipette delivers one fixed volume very accurately. A measuring cylinder is much less accurate.

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12 · Part 2

Paper 2 only | Acids, alkalis and titrations | [8]

Mark scheme

(a) **[2]**

- rough 23.40, titration 1 22.45, titration 2 22.85, titration 3 22.55 (cm^3)
- all four correct (2)
- two or three correct (1)

(b) **[2]**

- uses titrations 1 and 3 only (22.45 and 22.55) (1)
- mean titre = 22.50 cm^3 (1)

ALLOW 22.5 cm^3

ECF from (a)

(c) **[3]**

- amount of H_2SO_4 = 0.100 × 22.50 ÷ 1000 = 0.00225 mol (1)
- amount of KOH = 2 × 0.00225 = 0.00450 mol (1)
- concentration of KOH = 0.00450 ÷ 0.0250 = 0.180 mol/dm^3 (1)

ALLOW 0.18 mol/dm^3

ECF from the mean titre in (b)

ALLOW 0.0900 mol/dm^3 scores 2 marks (mole ratio not used)

(d) **[1]**

- the rough titration is done quickly to find the approximate end point, so it is likely to overshoot OR is less accurate (1)

ALLOW it is not concordant with the other titres

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Worked solution

(a) Titre = final reading − initial reading. Rough: 23.40 − 0.00 = 23.40. Titration 1: 22.95 − 0.50 = 22.45. Titration 2: 45.80 − 22.95 = 22.85. Titration 3: 23.75 − 1.20 = 22.55 cm^3.

(b) 22.45 and 22.55 differ by 0.10 cm^3, so they are concordant. 22.85 is 0.30 cm^3 or more away from both, so it is left out. Mean = (22.45 + 22.55) ÷ 2 = 22.50 cm^3.

(c) Change the volume to dm^3: 22.50 ÷ 1000 = 0.02250 dm^3. Amount of acid = 0.100 × 0.02250 = 0.00225 mol. The equation shows 2 mol KOH react with 1 mol H_2SO_4, so KOH = 2 × 0.00225 = 0.00450 mol. This was in 25.0 cm^3 = 0.0250 dm^3, so concentration = 0.00450 ÷ 0.0250 = 0.180 mol/dm^3. Check: 0.180 × 0.0250 = 0.00450 mol, half of which is 0.00225 mol, matching the acid.

(d) The rough titration is only a quick trial to find roughly where the end point is, so the acid is usually added too fast and the end point is overshot.

13 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [1]

Answer: C

Worked solution

Go through the solubility rules. Ammonium chloride: ammonium compounds are soluble. Calcium nitrate: all nitrates are soluble. Potassium carbonate: carbonates are insoluble except sodium, potassium and ammonium carbonates, so it is soluble. Barium sulfate is one of the exceptions to the rule that sulfates are soluble.

14 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [1]

Answer: B

Worked solution

Compare the formulae. NH_3 becomes NH_4^+, so ammonia has gained an H^+, a proton: it is the proton acceptor, the base. HCl has lost its H^+ and become Cl^-: it is the proton donor, the acid.

15 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [1]

Answer: D

Worked solution

Carbon dioxide only forms when a carbonate reacts with an acid, so the zinc compound must be zinc carbonate. To make a sulfate, the acid must be sulfuric acid. So the pair is zinc carbonate and dilute sulfuric acid.

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16 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [7]

Mark scheme

(a) **[3]**

- magnesium chloride; copper(II) sulfate; calcium nitrate; potassium sulfate
- all four correct (3)
- three correct (2)
- two correct (1)

ALLOW copper sulfate

(b) **[1]**

- fizzing OR bubbles OR effervescence OR the calcium carbonate disappears OR dissolves (1)

IGNORE carbon dioxide is made (not an observation)

(c) **[3]**

- 2 in front of HCl (1)
- $MgCl_2$ and H_2 as the products (1)
- $MgCl_2(aq)$ and $H_2(g)$ state symbols (1)
- $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
- the state symbol mark depends on the correct products

Worked solution

(a) Hydrochloric acid makes chlorides, sulfuric acid makes sulfates and nitric acid makes nitrates. The metal part of the name comes from the metal, base or carbonate: magnesium chloride, copper(II) sulfate, calcium nitrate and potassium sulfate.

(b) Calcium carbonate reacts with the acid to give off carbon dioxide gas, so you see fizzing, and the solid gets used up.

(c) Magnesium and hydrochloric acid give magnesium chloride and hydrogen. Mg^{2+} needs two Cl^- ions, so the formula is $MgCl_2$, and two HCl are needed: $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$. Magnesium chloride is dissolved (aq) and hydrogen is a gas (g).

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17 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [6]

Mark scheme

(a) **[1]**

- a base that is soluble in water (1)

ALLOW a soluble base OR a base that dissolves in water to form OH^- ions

(b) **[2]**

- sodium hydroxide (1)
- potassium hydroxide (1)
- deduct one mark for each extra incorrect substance given

(c) **[2]**

- ammonia accepts a proton OR H^+ (1)
- from the acid, which donates it, forming NH_4^+ (1)

ALLOW $NH_3 + H^+ \rightarrow NH_4^+$ for both marks

(d) **[1]**

- ammonium sulfate (1)

Worked solution

(a) Every alkali is a base, but only bases that dissolve in water are alkalis.

(b) All five substances are hydroxides, so all are bases. Common hydroxides are insoluble except those of sodium, potassium and calcium, so only sodium hydroxide and potassium hydroxide dissolve and are alkalis.

(c) A base is a proton acceptor. NH_3 becomes NH_4^+, so it has gained one H^+ (a proton) from the acid.

(d) Ammonia forms ammonium salts. Sulfuric acid forms sulfates, so the salt is ammonium sulfate.

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18 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [7]

Mark scheme

(a) **[1]**

- $CuSO_4 + H_2O$ (1)

IGNORE state symbols

(b) **[1]**

- to speed up the reaction (1)

ALLOW so the copper(II) oxide reacts OR dissolves faster

(c) **[1]**

- to make sure all of the acid reacts OR is used up (1)

ALLOW so that no acid is left in the solution

(d) **[4]**

Indicative points, any four (1 each):

- filter the mixture to remove the excess (unreacted) copper(II) oxide (1)
- heat the filtrate in the evaporating basin to evaporate some of the water OR until the crystallisation point is reached (1)

ALLOW until crystals form on a glass rod dipped in the solution and taken out to cool

- leave the solution to cool so that crystals form (1)
- filter off OR pour off the liquid to separate the crystals (1)
- dry the crystals by pressing them between filter papers OR by leaving them in a warm place (1)

REJECT heating until all the water has gone (for the evaporation mark)

Worked solution

(a) A base and an acid react to give a salt and water: $CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$.

(b) Warm acid reacts faster with the copper(II) oxide.

(c) If the copper(II) oxide is in excess, all of the acid is used up, so the solution contains only copper(II) sulfate and water. The leftover solid is easy to remove.

(d) Filter to remove the unreacted black copper(II) oxide; the blue filtrate is copper(II) sulfate solution. Heat it in an evaporating basin until it is concentrated (crystals start to appear on a cooled glass rod), but do not boil it dry. Leave it to cool so blue crystals grow. Separate the crystals from the remaining solution and dry them between filter papers or in a warm place. Strong heating would drive off the water of crystallisation.

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19 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [6]

Mark scheme

(a) **[1]**

- zinc oxide (1)

(b) **[2]**

- $ZnCl_2$ + H_2O + CO_2 as products (1)
- $ZnCl_2(aq)$, $H_2O(l)$ and $CO_2(g)$ state symbols (1)
- $ZnCO_3(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2O(l) + CO_2(g)$
- the state symbol mark depends on the correct products

(c) **[1]**

- some solid zinc carbonate remains (does not dissolve) OR the fizzing stops while solid is left (1)

(d)(i) **[1]**

- copper does not react with dilute hydrochloric acid OR copper is too unreactive (1)

ALLOW copper is less reactive than hydrogen

(d)(ii) **[1]**

- copper(II) oxide OR copper(II) hydroxide OR copper(II) carbonate (1)

ALLOW copper oxide OR copper hydroxide OR copper carbonate

Worked solution

(a) Zinc and acid give zinc chloride and hydrogen. Zinc carbonate gives zinc chloride, water and carbon dioxide. Zinc oxide is a base, and a base and an acid give only a salt and water.

(b) $ZnCO_3(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2O(l) + CO_2(g)$. Zinc chloride is dissolved, water is a liquid and carbon dioxide is a gas.

(c) When all the acid has been used up, the extra zinc carbonate has nothing to react with, so it stays as a solid and the fizzing stops.

(d) Copper is very low in the reactivity series and does not react with dilute acids. A copper compound that is a base or a carbonate, such as copper(II) oxide or copper(II) carbonate, does react with the acid to make copper(II) chloride.

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20 · Part 2

Paper 2 only | Acids, bases and salt preparations | [7]

Mark scheme

(a) **[2]**

- sodium hydroxide is soluble OR is an alkali (1)
- so any excess stays dissolved in the solution and cannot be removed by filtering, so the salt would be impure (1)

(b) **[5]**

Indicative points, any five (1 each):

- use the pipette to put 25.0 cm^3 of sodium hydroxide solution into the conical flask (1)
- add a few drops of methyl orange (1)
- add sulfuric acid from the burette, swirling, until the indicator just changes colour (from yellow to orange OR red) (1)
- record the volume of acid added (1)
- repeat with the same volumes of acid and sodium hydroxide solution but without the indicator (1)
- heat the solution in the evaporating basin to evaporate some of the water OR to the crystallisation point (1)
- leave to cool so that crystals form (1)
- filter off the crystals and dry them between filter papers OR in a warm place (1)

Worked solution

(a) With an insoluble base, the extra solid sits in the flask and is easily filtered off. Sodium hydroxide is soluble, so any extra just stays dissolved and passes through the filter paper. The sodium sulfate would then be mixed with sodium hydroxide.

(b) Ben must find exactly how much acid neutralises the alkali. He pipettes 25.0 cm^3 of sodium hydroxide solution into a conical flask, adds methyl orange and runs in acid from the burette until the colour just changes from yellow to orange. He notes the volume. The indicator would contaminate the salt, so he mixes the same two volumes again without indicator. Now the solution contains only sodium sulfate and water. He evaporates some of the water in an evaporating basin, leaves the solution to cool and crystallise, then filters off the crystals and dries them.

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21 · Part 2

Paper 2 only | Acids, bases and salt preparations | [7]

Mark scheme

(a) **[1]**

- a white precipitate forms (1)

> **ALLOW** white solid forms OR mixture turns cloudy white

(b) **[2]**

- $PbSO_4(s)$ (1)
- $2NaNO_3(aq)$ (1)
- $Pb(NO_3)_2(aq) + Na_2SO_4(aq) \rightarrow PbSO_4(s) + 2NaNO_3(aq)$

(c) **[4]**

Indicative points, any four (1 each):

- filter the mixture (1)
- the lead(II) sulfate stays on the filter paper as the residue (1)
- wash the residue with distilled water (1)
- to remove soluble impurities such as sodium nitrate (solution) (1)
- dry the solid in an oven (at a low temperature) OR in a warm place OR between filter papers (1)

Worked solution

(a) Lead(II) sulfate is insoluble, so it appears as a white solid (precipitate) as soon as the solutions are mixed.

(b) The ions swap partners. Lead(II) sulfate is the solid; sodium nitrate stays dissolved. Two NO_3^- ions come from $Pb(NO_3)_2$, so $2NaNO_3$ form: $Pb(NO_3)_2(aq) + Na_2SO_4(aq) \rightarrow PbSO_4(s) + 2NaNO_3(aq)$.

(c) Pour the mixture through filter paper. The solid lead(II) sulfate is caught on the paper and the sodium nitrate solution passes through. Some solution still wets the solid, so wash it with distilled water to remove the dissolved sodium nitrate. Then dry the solid in a warm oven.

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22 · Part 2

Paper 2 only | Acids, bases and salt preparations | [7]

Mark scheme

(a) **[1]**

- calcium carbonate is insoluble in water, so it cannot be used as a solution (1)

ALLOW calcium carbonate is insoluble

(b) **[3]**

- amount of $BaCO_3$ = 0.0120 mol (1 : 1 ratio) (1)
- Mr of $BaCO_3$ = 137 + 12 + (3 × 16) = 197 (1)
- mass = 0.0120 × 197 = 2.36 g (1)

ALLOW 2.364 g OR 2.4 g

ECF from an incorrect Mr

(c) **[2]**

- percentage yield = 2.05 ÷ 2.364 × 100 (1)
- = 86.7% (1)

ALLOW any value from 86.7% to 86.9% OR 87%

ECF from (b)

(d) **[1]**

- some barium carbonate was lost during filtering OR was left on the filter paper OR in the beaker OR passed through the filter paper OR was lost when washing (1)

REJECT the product was wet (this would increase the mass)

Worked solution

(a) This method mixes two solutions. Calcium carbonate is insoluble, so it cannot be made into a solution to provide carbonate ions.

(b) The equation shows 1 mol $BaCl_2$ gives 1 mol $BaCO_3$, so 0.0120 mol of $BaCO_3$ can form. Mr = 137 + 12 + 3 × 16 = 197. Mass = 0.0120 × 197 = 2.364 g, which is 2.36 g to 3 significant figures.

(c) Percentage yield = actual yield ÷ theoretical yield × 100 = 2.05 ÷ 2.364 × 100 = 86.7%. Check: 86.7% of 2.364 g is 2.05 g.

(d) Some of the solid is always lost: it sticks to the beaker or filter paper, or a little passes through the filter paper.

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23 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [6]

Mark scheme

(a) **[3]**

- potassium carbonate soluble; zinc nitrate soluble; silver chloride insoluble; calcium sulfate insoluble; magnesium hydroxide insoluble; ammonium carbonate soluble
- all six correct (3)
- four or five correct (2)
- two or three correct (1)

(b) **[2]**

- calcium sulfate forms, which is insoluble (1)
- it forms a layer on the calcium carbonate that stops the acid reaching it (1)

(c) **[1]**

- calcium chloride is soluble, so it dissolves and does not coat the calcium carbonate (1)

Worked solution

(a) Potassium and ammonium compounds are soluble, even their carbonates. All nitrates are soluble. Silver chloride is an exception to the soluble chlorides. Calcium sulfate is an exception to the soluble sulfates. Magnesium hydroxide is insoluble, since only sodium, potassium and calcium hydroxides dissolve.

(b) The reaction makes calcium sulfate. It is insoluble, so it stays on the surface of the lumps as a solid coating. The acid can no longer touch the calcium carbonate underneath, so the fizzing stops.

(c) With hydrochloric acid the salt is calcium chloride, which is soluble. It dissolves as soon as it forms, so the surface of the calcium carbonate stays exposed and the reaction continues.

24 · Part 2

Paper 1 & 2 | Chemical tests | [1]

Answer: A

Worked solution

Learn the three halide results: Cl^- white, Br^- cream, I^- yellow. A cream precipitate therefore means bromide ions.

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25 · Part 2 Paper 1 & 2 Chemical tests [1]

Answer: C

Worked solution

Sodium hydroxide solution results: Cu^{2+} blue precipitate, Fe^{2+} green precipitate, Fe^{3+} orange-brown precipitate. Green means Fe^{2+}.

26 · Part 2 Paper 1 & 2 Chemical tests [1]

Answer: D

Worked solution

The copper(II) sulfate test shows water is present but cannot tell pure from impure water. To show purity, use a physical test: a pure substance has a sharp, fixed boiling point. Pure water boils at exactly 100 °C.

27 · Part 2 Paper 1 & 2 Chemical tests [6]

Mark scheme

(a) **[3]**

- oxygen: a glowing splint relights (1)
- hydrogen: a lighted splint gives a (squeaky) pop (1)
- chlorine: damp (blue) litmus paper is bleached OR turns white (1)

> **ALLOW** damp blue litmus paper turns red then white

- each mark needs both the test and the result

(b) **[2]**

- bubble the gas through limewater (1)
- the limewater turns milky OR cloudy (1)

(c) **[1]**

- carbonate ions OR CO_3^{2-} are present (1)

Worked solution

(a) Oxygen helps things burn, so a glowing splint bursts back into flame. Hydrogen burns with a squeaky pop when a lighted splint is held at the mouth of the tube. Chlorine bleaches damp litmus paper white.

(b) Carbon dioxide turns limewater milky (cloudy white).

(c) A solid that fizzes with dilute acid and gives carbon dioxide contains carbonate ions.

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28 · Part 2 Paper 1 & 2 Chemical tests [6]

Mark scheme

(a) **[3]**

- clean a (nichrome OR platinum) wire loop by dipping it in (concentrated) hydrochloric acid and holding it in a flame until it gives no colour (1)
- dip the loop in hydrochloric acid and then in the solid sample (1)
- hold the loop in a hot (blue OR roaring) Bunsen burner flame and observe the colour of the flame (1)

(b) **[3]**

- 1 Li^+, 2 K^+, 3 Ca^{2+}, 4 Cu^{2+}
- all four correct (3)
- three correct (2)
- two correct (1)

ALLOW names of the ions OR of the metals

Worked solution

(a) First clean the wire loop so no colour from an earlier sample is left: dip it in hydrochloric acid and heat it until the flame shows no colour. Moisten the loop with acid so the solid sticks, dip it in the sample and hold it in the hottest part of a blue Bunsen flame. Note the colour.

(b) Red is lithium, Li^+. Lilac is potassium, K^+. Orange-red is calcium, Ca^{2+}. Blue-green is copper(II), Cu^{2+}.

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29 · Part 2 Paper 1 & 2 Chemical tests [7]

Mark scheme

(a) **[3]**

- X copper(II) chloride (1)
- Y sodium iodide (1)
- Z iron(III) bromide (1)

ALLOW copper chloride

REJECT iron(II) bromide

(b) **[1]**

- $Cu(OH)_2$ (1)

(c) **[2]**

- iron(II) ions would give a green precipitate (with sodium hydroxide solution) (1)
- the orange-brown precipitate shows that iron(III) ions are present (1)

(d) **[1]**

- to remove carbonate ions, which would also give a precipitate with silver nitrate (1)

ALLOW to stop other ions (such as carbonate or hydroxide) giving a precipitate OR to prevent a false positive result

Worked solution

(a) X: a blue precipitate with sodium hydroxide means Cu^{2+}, and a white precipitate with acidified silver nitrate means Cl^-, so X is copper(II) chloride. Y: no precipitate with sodium hydroxide and a yellow flame means Na^+; a yellow precipitate with silver nitrate means I^-, so Y is sodium iodide. Z: an orange-brown precipitate means Fe^{3+} and a cream precipitate means Br^-, so Z is iron(III) bromide.

(b) Cu^{2+} combines with two OH^- ions, so the blue precipitate is copper(II) hydroxide, $Cu(OH)_2$.

(c) Iron(II) ions give a green precipitate. Sofia saw orange-brown, which is the result for iron(III) ions.

(d) Carbonate ions also form a precipitate with silver ions. The nitric acid reacts with any carbonate ions first, so a precipitate can only come from a halide ion.

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30 · Part 2 Paper 1 & 2 Chemical tests [7]

Mark scheme

(a) **[3]**

- add sodium hydroxide solution and warm (1)
- test the gas with damp red litmus paper (1)
- the litmus paper turns blue (1)

ALLOW damp universal indicator paper turns blue OR purple

(b) **[2]**

- add dilute hydrochloric acid and then barium chloride solution (1)
- a white precipitate forms (1)

ALLOW acidified barium chloride solution

REJECT sulfuric acid as the acid

(c)(i) **[1]**

- potassium nitrate contains no ammonium ions, so no ammonia (gas) is given off (1)

(c)(ii) **[1]**

- potassium hydroxide is soluble, so no precipitate forms (1)

ALLOW there is no visible change OR no precipitate forms

Worked solution

(a) Ammonium ions react with warm sodium hydroxide solution to give ammonia gas. Ammonia is alkaline, so it turns damp red litmus paper blue.

(b) Add dilute hydrochloric acid, then barium chloride solution. Sulfate ions form insoluble barium sulfate, a white precipitate.

(c)(i) The test in (a) works only because ammonium ions make ammonia. Potassium nitrate has no ammonium ions, so no ammonia forms and the litmus stays red.

(c)(ii) The sodium hydroxide test works by forming an insoluble metal hydroxide precipitate. Potassium hydroxide is soluble, so nothing is seen. A flame test (lilac flame) is used to identify potassium ions instead.

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